

Axilla (Armpit)

Organised anatomy notes with source-PDF diagrams - spacious Avenir Next LT Pro layout

Core idea

The axilla is a fat-filled pyramidal space between the upper limb and the lateral thoracic wall. It works as a passage for the axillary vessels, cords of the brachial plexus and lymphatics between the neck, thorax and upper limb.

Shape	Main contents	Most clinical region	Exam focus
Truncated four-sided pyramid: apex, base and four walls	Axillary artery and vein, cords of brachial plexus, lymph nodes, fat, axillary tail of breast	Floor/base of axilla and axillary lymph nodes	Boundaries, artery branches, lymph nodes, brachial plexus and applied anatomy

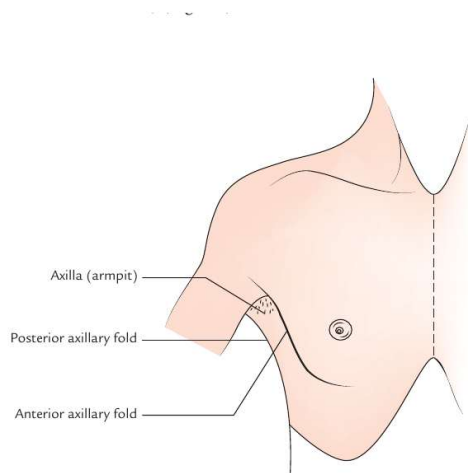


Fig. 4.1 Location of the axilla.

Source diagram: location of the axilla.

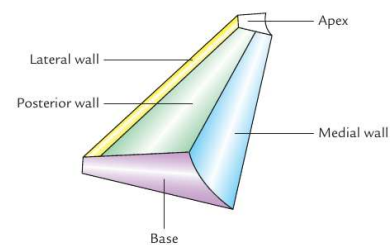


Fig. 4.2 Boundaries of the axilla (Note anterior wall is not seen).

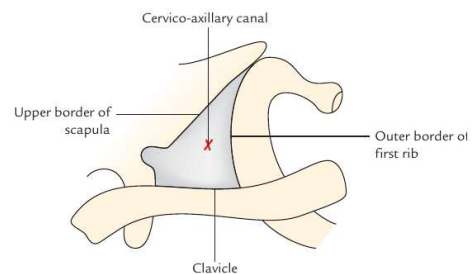


Fig. 4.3 Boundaries of the cervico-axillary canal (apex of the axilla).

Source diagram: pyramidal shape and cervico-axillary canal.

Clinically, the axilla is important because its lymph nodes are routinely palpated during breast examination, abscesses may occur in this region, and the neurovascular structures are vulnerable when the arm is abducted.

1. BOUNDARIES OF THE AXILLA

The axilla resembles a truncated pyramid. It has an apex, a base or floor, and four walls. The apex is also called the cervico-axillary canal because the axillary artery and brachial plexus enter the axilla through it from the neck.

Apex / cervico-axillary canal

- Directed upwards and medially into the root of the neck.
- Bounded anteriorly by clavicle, posteriorly by upper border of scapula, and medially by outer border of first rib.
- Transmits axillary artery and brachial plexus from neck to axilla.

Base / floor

- Formed by axillary fascia at the lower end of axilla.
- Bounded anteriorly by anterior axillary fold, posteriorly by posterior axillary fold, and medially by lateral chest wall.
- Most dependent part; important site for drainage of axillary abscess.

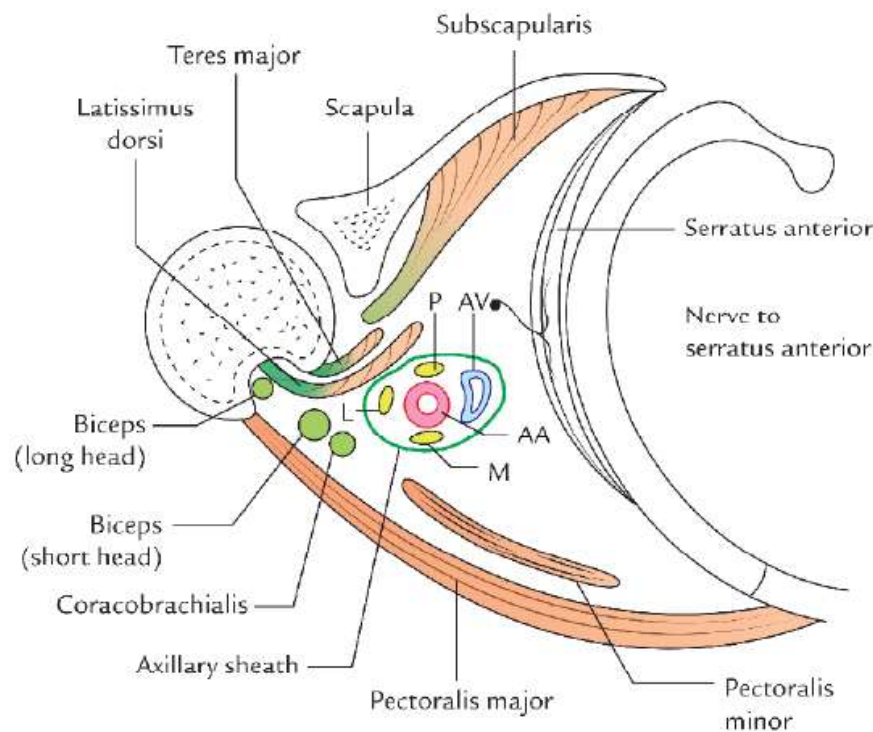


Fig. 4.4 Boundaries and contents of the axilla as seen in horizontal section (AV = axillary vein, AA = axillary artery)

Source diagram: horizontal section showing boundaries and contents of the axilla.

Anterior and posterior walls

- Anterior wall: pectoralis major, subclavius and pectoralis minor.
- Posterior wall: subscapularis above; latissimus dorsi and teres major below.
- Anterior and posterior axillary folds are surface landmarks.

Medial and lateral walls

- Medial wall: upper four or five ribs and intercostal spaces covered by serratus anterior.
- Lateral wall: narrow wall formed by bicipital groove of humerus with coracobrachialis and short head of biceps.
- The lateral wall is narrow because anterior and

posterior walls converge at this site.

2. CONTENTS OF THE AXILLA

The contents are embedded in axillary fat. The fat cushions vessels and nerves, but these structures become exposed to traction and compression during marked abduction of the arm.

Content	Important point
Axillary artery and branches	Main arterial supply pathway to the upper limb; divided into three parts by pectoralis minor.
Axillary vein and tributaries	Lies medial to axillary artery; becomes subclavian vein at outer border of first rib.
Cords of brachial plexus	Lateral, medial and posterior cords are named according to their relation to the second part of the axillary artery.
Axillary lymph nodes	20-30 nodes in pectoral, subscapular, lateral, central and apical groups.

Other contents

- Fibrofatty tissue.
- Axillary tail of breast.
- Long thoracic nerve.
- Intercostobrachial nerve.

Protection and vulnerability

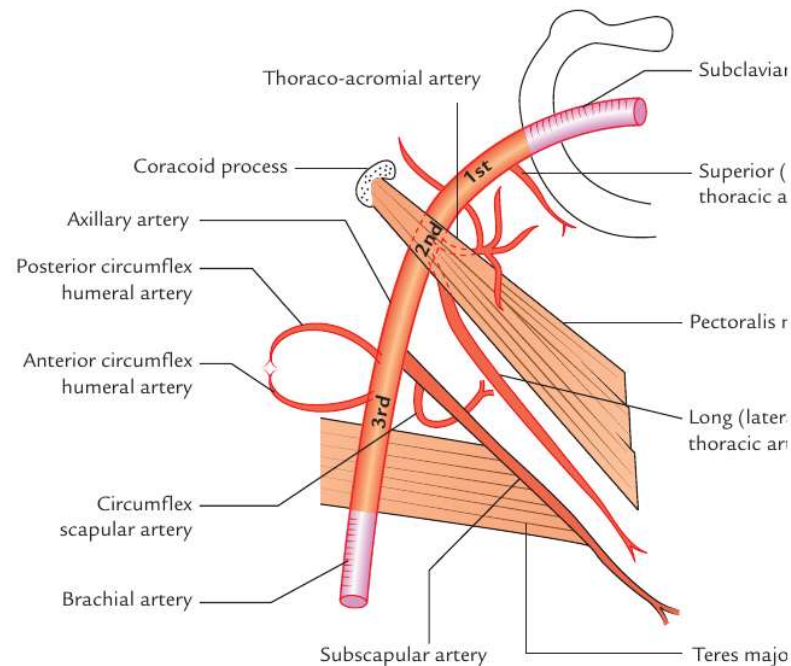
- Structures are normally protected by the arm and axillary fat.
- They may be injured when the arm is abducted.
- Clinical examination should be gentle and systematic.

Surface landmark

The perpendicular line dropped from a point midway between the anterior and posterior axillary folds is called the midaxillary line. It is an important surface landmark for thoracic and axillary anatomy.

3. AXILLARY ARTERY

The axillary artery is the main artery of the upper limb. It begins at the outer border of the first rib as the continuation of the subclavian artery and ends at the lower border of teres major by becoming the brachial artery.



Source diagram: course and branches of the axillary artery.

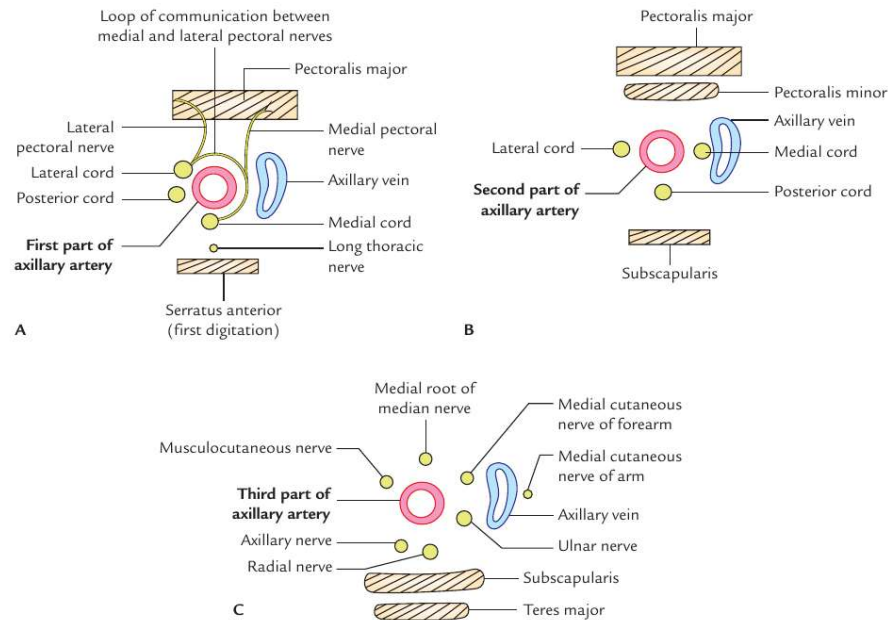
Part	Position relative to pectoralis minor	Branches
First part	Superior/proximal to pectoralis minor.	Superior thoracic artery.
Second part	Posterior/deep to pectoralis minor.	Thoraco-acromial artery and lateral thoracic artery.
Third part	Inferior/distal to pectoralis minor.	Subscapular artery, anterior circumflex humeral artery and posterior circumflex humeral artery.

Quick memory

Axillary artery gives 1 branch from the first part, 2 branches from the second part and 3 branches from the third part. Remember the sequence: Superior thoracic - Thoraco-acromial, Lateral thoracic - Subscapular, Anterior circumflex humeral, Posterior circumflex humeral.

4. RELATIONS OF THE AXILLARY ARTERY

The axillary vein lies medial to the axillary artery throughout its course. The cords of the brachial plexus are arranged around the second part of the artery: lateral cord lateral, medial cord medial and posterior cord posterior.



Source diagram: relations of first, second and third parts of the axillary artery.

Part	Anterior	Posterior	Medial / lateral key relations
First	Pectoralis major; loop between medial and lateral pectoral nerves.	Medial cord, long thoracic nerve, serratus anterior.	Medial: axillary vein. Lateral: lateral and posterior cords.
Second	Pectoralis minor.	Posterior cord and subscapularis.	Medial: medial cord and axillary vein. Lateral: lateral cord.
Third	Medial root of median nerve.	Radial nerve, axillary nerve, subscapularis and teres major.	Medial: axillary vein, medial cutaneous nerve of forearm, ulnar nerve. Lateral: musculocutaneous nerve.

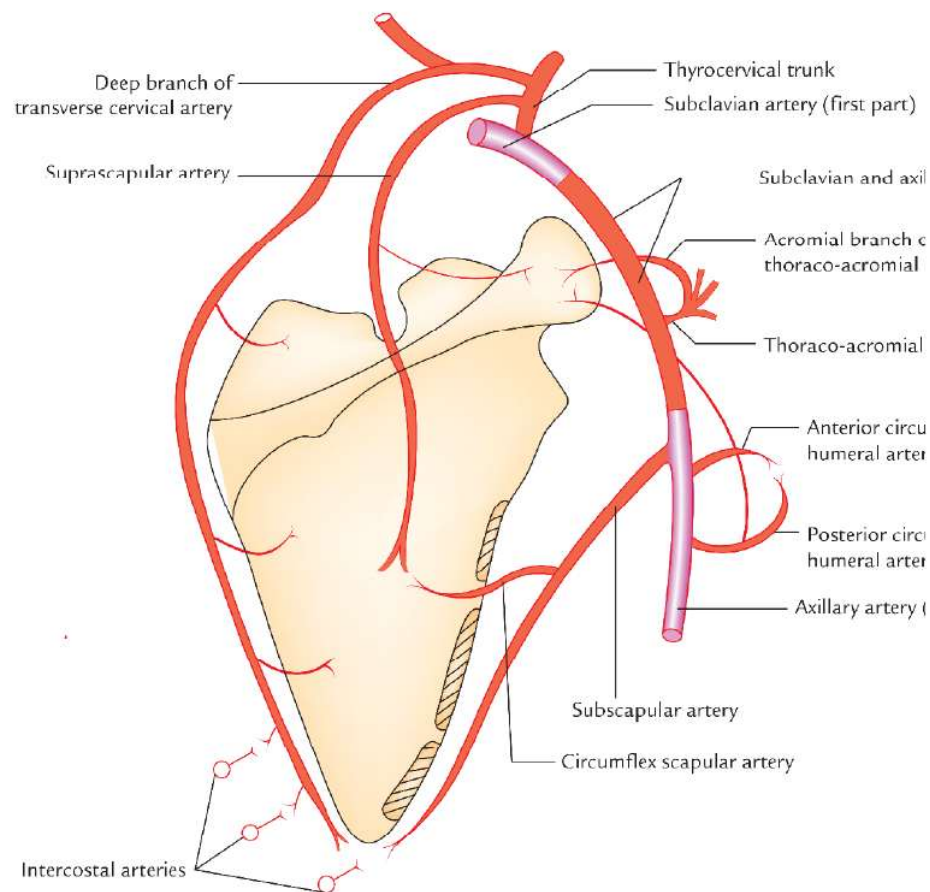
5. AXILLARY VEIN AND SCAPULAR ANASTOMOSIS

Axillary vein

- Formed at the lower border of teres major by union of basilic vein and venae comitantes of brachial artery.
- Runs along the medial side of the axillary artery.
- Ends at the outer border of first rib by becoming the subclavian vein.
- Tributaries correspond to branches of axillary artery; cephalic vein joins after piercing clavipectoral fascia.

Clinical note

- An axillary arch muscle may overlie the vein.
- Unaccustomed overhead movements can compress the vein and may cause spontaneous thrombosis.
- The axillary vein has little sheath, so it can expand during increased venous return.



Source diagram: scapular anastomosis around the scapula.

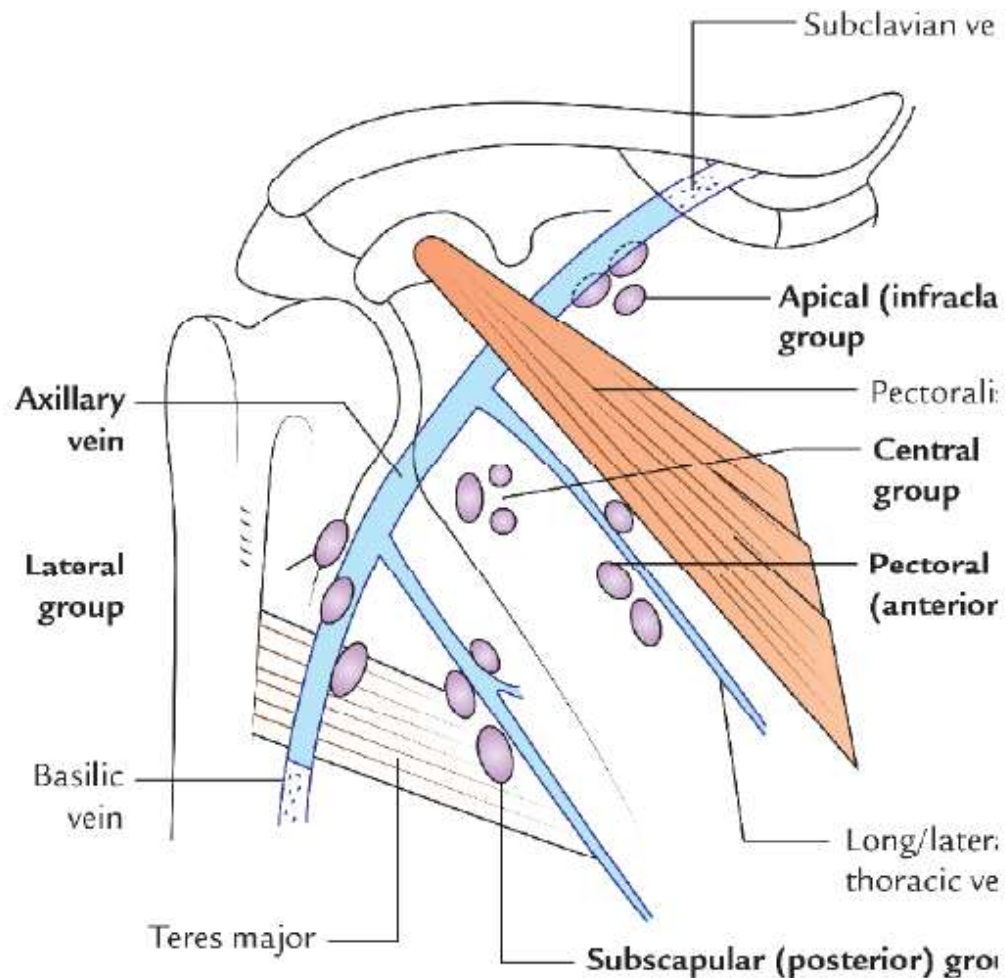
Site	Arteries taking part
Around body of scapula	Suprascapular artery, circumflex scapular artery and deep branch of transverse cervical artery.
Over acromion process	Acromial branches of thoraco-acromial, suprascapular and posterior circumflex humeral arteries.

Collateral circulation

If the subclavian or axillary artery is blocked between the first part of subclavian artery and third part of axillary artery, scapular anastomosis may provide a collateral route to the upper limb.

6. AXILLARY LYMPH NODES

The axillary lymph nodes lie in the fibrofatty tissue of the axilla. Their number varies from 20 to 30. They are important because much of the lymph from the breast drains into them and they are routinely examined in suspected breast disease.



Golden Facts to Remember

- | | |
|---|---|
| ▶ Most dependant part of the axilla | Floor of the axilla |
| ▶ Axillary lymph node group receiving most of the lymph from the breast | Anterior or pectoral group |
| ▶ Largest branch of the brachial plexus | Radial nerve |
| ▶ Kuntz's nerve | Communicating branch between T1 and T2 nerves, which carry sympathetic fibres from 3rd thoracic ganglion to the upper limb via T2 nerve |
| ▶ Cervico-axillary canal | Truncated apex of the axilla through which the axillary artery and brachial plexus enter the axilla from neck |

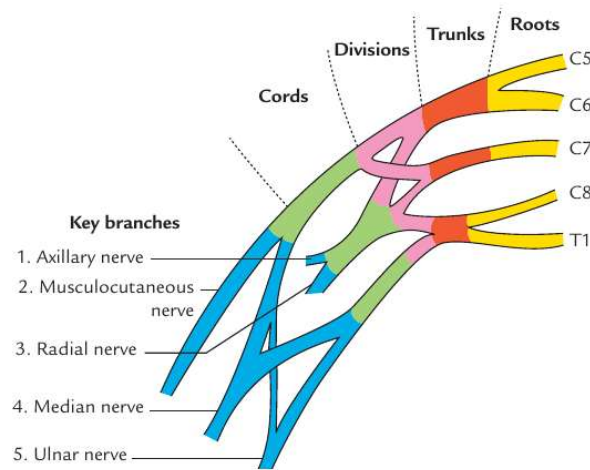
Group	Position	Drainage / importance
Anterior / pectoral	Along lateral thoracic vein at lower border of pectoralis minor.	Upper half of trunk anteriorly and major part of breast; related to axillary tail of Spence.
Posterior / subscapular	On posterior axillary fold along subscapular vein.	Upper half of trunk posteriorly and axillary tail of breast.
Lateral	Along upper part of humerus in relation to axillary vein.	Drains lymph from upper limb.
Central	In upper part of axilla.	Receives from other groups and drains into apical group; intercostobrachial nerve passes among these nodes.
Apical / infraclavicular	Deep to clavipectoral fascia at apex of axilla along axillary vein.	Receives directly from upper breast and indirectly from rest of breast; drains to subclavian lymph trunk/right side and thoracic duct/left side.

Level system in breast surgery

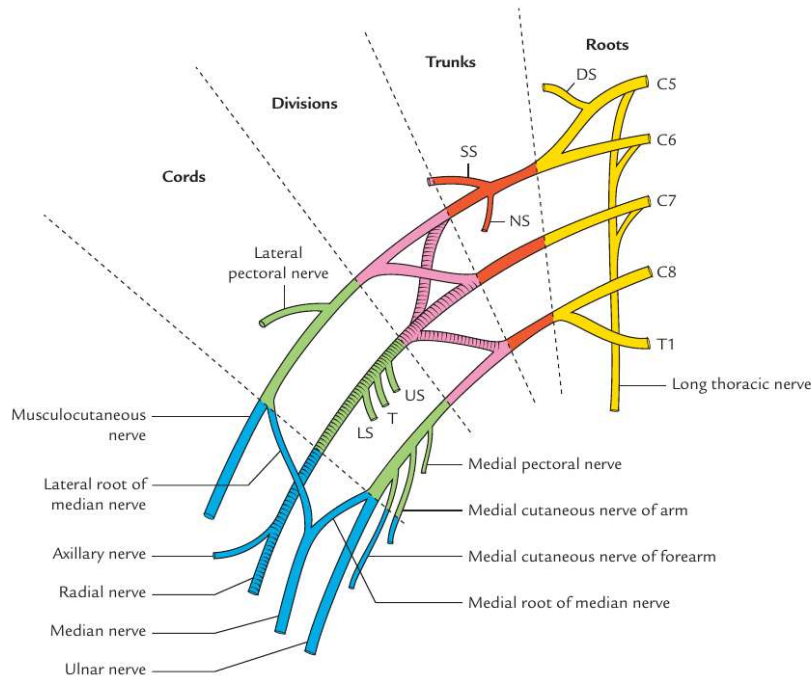
Level I nodes lie lateral to lower border of pectoralis minor, Level II nodes lie deep to pectoralis minor, and Level III nodes lie medial to upper border of pectoralis minor. Sentinel lymph nodes are usually Level I nodes.

7. BRACHIAL PLEXUS - FORMATION AND COMPONENTS

The brachial plexus is formed by anterior rami of C5, C6, C7, C8 and T1 spinal nerves, with small possible contributions from C4 to T2. Roots and trunks are in the neck, divisions are behind the clavicle, and cords are in the axilla.



Source diagram: components and key branches of the brachial plexus.



Source diagram: brachial plexus and its branches.

Component	Formation / position
Roots	Anterior primary rami of C5 to T1; located in the neck deep to scalenus anterior.
Trunks	Upper trunk = C5 + C6; middle trunk = C7; lower trunk = C8 + T1. They lie in the neck between scalenus medius and scalenus anterior.
Divisions	Each trunk divides into anterior and posterior divisions. They lie behind the clavicle.
Cords	Lateral cord from anterior divisions of upper and middle trunks; medial cord from anterior division of lower trunk; posterior cord from posterior divisions of all three trunks.

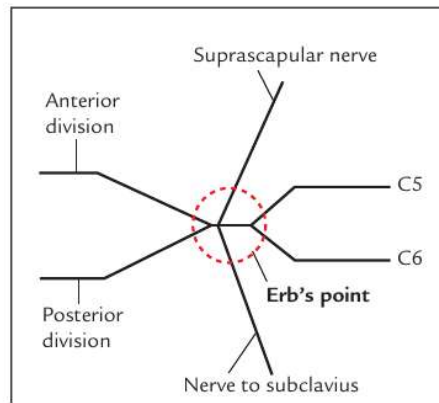
Prefixed and postfixed plexus

If contribution from C4 is large and T2 is absent, it is called prefixed brachial plexus. If contribution from T2 is large and C4 is absent, it is called postfixed brachial plexus.

8. BRANCHES OF BRACHIAL PLEXUS

Branches from roots and trunks are supraclavicular branches, while branches from cords are infraclavicular branches. The largest terminal branch is the radial nerve.

Origin	Branches	Root values / notes
Roots	Long thoracic nerve; dorsal scapular nerve.	Long thoracic: C5-C7. Dorsal scapular: C5.
Trunks	Suprascapular nerve; nerve to subclavius.	Both mainly C5-C6.
Lateral cord	Lateral pectoral nerve; lateral root of median nerve; musculocutaneous nerve.	Mainly C5-C7.
Medial cord	Medial pectoral nerve; medial cutaneous nerve of arm; medial cutaneous nerve of forearm; medial root of median nerve; ulnar nerve.	Mainly C8-T1; ulnar nerve also receives C7.
Posterior cord	Radial nerve; axillary nerve; thoracodorsal nerve; upper and lower subscapular nerves.	Radial: C5-T1. Axillary, upper and lower subscapular: C5-C6. Thoracodorsal: C6-C8.



Source diagram: Erb's point at the upper trunk of brachial plexus.

Erb point

Erb point is the region of the upper trunk where six nerves meet: C5 and C6 roots join to form the upper trunk; the upper trunk gives suprascapular nerve and nerve to subclavius, then divides into anterior and posterior divisions.

9. APPLIED ANATOMY - LYMPH NODES AND ABSCESS

Palpation of axillary nodes

- Part of clinical breast examination.
- Cancer breast can involve axillary lymph nodes.
- Central nodes may compress intercostobrachial nerve causing pain along the inner arm.
- Axillary tail of breast may be confused with enlarged lymph nodes.

Axillary abscess

- Arises from infection and suppuration of axillary lymph nodes.
- May enlarge considerably before pain is felt.
- Pus may track into neck/arm via axillary sheath or between pectoral muscles through clavipectoral fascia.
- Drained through floor of axilla, nearer medial wall, midway between anterior and posterior axillary folds.

10. APPLIED ANATOMY - BRACHIAL PLEXUS LESIONS

Brachial plexus lesions are understood by linking spinal segments to movements: C5 controls shoulder adduction, C6-C7 shoulder abduction, C5-C6 elbow flexion, C6-C7 elbow extension, and C8-T1 flexion of wrist and fingers.

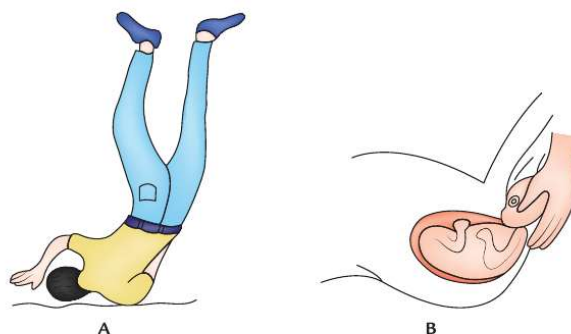


Fig. 4.12 Injury of the upper brachial plexus leading to excessive increase in the angle between the head and shoulder: **A**, fall from the height and landing on a shoulder; **B**, Traction of the arm and hyperextension of the neck.

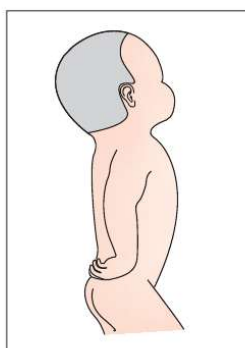


Fig. 4.13 Policeman receiving a tip position of the upper limb in Erb's paralysis.

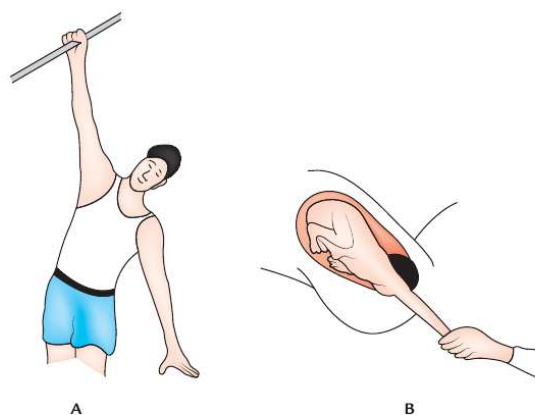


Fig. 4.14 Injury of the lower brachial plexus leading to excessive increase in the angle between the trunk and shoulder: **A**, sudden upward pull of the arm; **B**, arm pulled during delivery.

Source diagrams: mechanisms and positions in upper and lower brachial plexus injuries.

11. ERB'S PARALYSIS AND KLUMPKE'S PARALYSIS

Feature	Erb's paralysis	Klumpke's paralysis
Roots involved	C5 and C6.	C8 and T1.

Mechanism	Excessive increase in angle between head and shoulder; fall on shoulder or traction during birth.	Hyperabduction; fall on outstretched hand, arm pulled into machinery or traction during delivery.
Muscles affected	Deltoid, supraspinatus, infraspinatus, biceps, brachialis, brachioradialis, supinator and extensor carpi radialis longus.	All intrinsic muscles of the hand, with involvement of flexors of wrist and fingers.
Limb position	Policeman/porter/waiter's tip: arm adducted, medially rotated; elbow extended; forearm pronated.	Claw hand.
Sensory/autonomic	Minimal sensory loss along outer aspect of arm; no autonomic signs.	Sensory loss along medial forearm and hand; Horner syndrome may occur.

Surgical approach to axilla

During excision of axillary lymph nodes for breast cancer, structures at risk are intercostobrachial nerve, long thoracic nerve, thoracodorsal nerve and thoracodorsal artery. These should be protected.

12. GOLDEN FACTS AND FINAL REVISION SHEET

Golden fact	Answer
Most dependent part of axilla	Floor/base of the axilla.
Axillary node group receiving most lymph from breast	Anterior or pectoral group.
Largest branch of brachial plexus	Radial nerve.
Kuntz nerve	Communicating branch between T1 and T2 nerves carrying sympathetic fibres from 3rd thoracic ganglion to upper limb via T2 nerve.
Cervico-axillary canal	Truncated apex of axilla through which axillary artery and brachial plexus enter axilla from neck.

Clinical case pattern

A baby with right upper limb adducted and medially rotated, forearm extended and pronated after forceps delivery has policeman/porter tip position due to Erb's paralysis at Erb point involving C5-C6 roots.

One-minute boundaries - Apex: clavicle, upper border of scapula, outer border of first rib. - Base: axillary fascia/floor. - Anterior wall: pectoral muscles and subclavius. - Posterior wall: subscapularis, latissimus dorsi,	One-minute contents - Axillary artery and branches. - Axillary vein and tributaries. - Cords of brachial plexus. - Axillary lymph nodes, fat, axillary tail of breast, long thoracic and intercostobrachial nerves.
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teres major.	
One-minute clinical • Palpate axillary nodes in breast examination. • Drain abscess through floor of axilla near medial wall. • Protect intercostobrachial, long thoracic and thoracodorsal nerves in axillary surgery. • Differentiate Erb's and Klumpke's palsy by roots and limb position.	Memory hooks • Axillary artery branches: 1-2-3 from first-second-third parts. • Cords named around the second part of artery. • Level I nodes are usually sentinel nodes. • Radial nerve is the largest branch of brachial plexus.

A baby boy was delivered in the hospital by an obstetrician by pulling baby's head using forceps (*forceps delivery*). Two weeks later the parents took the baby to the pediatrician for check-up. While examining the baby, the pediatrician found that baby's right arm was medially rotated and adducted while his forearm was extended and pronated. He also noticed sensory loss on the lateral aspect of the right upper limb.

Questions

1. Name the position of upper limb of baby observed by the pediatrician.
2. Which position of the limb is characteristic of this clinical condition?

3. Name the site of lesion and the cause that produced this condition.
4. What is the cause of sensory loss in the upper limb?

Answers

1. Policeman's tip hand/Porter's tip hand.
2. Erb's paralysis.
3. Erb's point due to excessive separation of neck and shoulder caused by pulling the baby's head during delivery.
4. Involvement of ventral root of C5 and C6 spinal nerves.

Source clinical case study: Erb's paralysis after forceps delivery.